THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 5 & 6: **Number Play, We Distribute, Yet Things Multiply** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Is 8109 divisible by 9? 1

(a) Only if the last digit is even **(b)** Yes

(c) No **(d)** Only if the number is prime

2. Is 2728 divisible by 11? 1

(a) Only if it ends in 1 **(b)** Yes

(c) Only if it is odd **(d)** No

3. The smallest digit that replaces the blank in 47_5 so that the number is divisible by 3 is: 1

(a) 3 **(b)** 2 **(c)** 7 **(d)** 4

4. The smallest digit that replaces the blank in 2_43 so that the number is divisible by 9 is: 1

(a) 2 **(b)** 1 **(c)** 5 **(d)** 0

5. The LCM of 15 and 25 is: 1

(a) 375 **(b)** 90 **(c)** 75 **(d)** 5

6. The value of 99×25 is: 1

(a) 2500 **(b)** 2475 **(c)** 2450 **(d)** 2485

7. The value of 31^2 is: 1

(a) 930 **(b)** 992 **(c)** 961 **(d)** 971

8. The value of 103^2 is: 1

(a) 10506 **(b)** 10619 **(c)** 10609 **(d)** 10712

9. The value of $105^2 - 95^2$ is: 1

(a) 20050 **(b)** 2000 **(c)** 100 **(d)** 40000

10. If $a = 2$ and $b = 7$, then $(a + b)^2$ equals: 1

(a) 25 **(b)** 53 **(c)** 67 **(d)** 81

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the Kaprekar process for 3-digit numbers briefly. 2

12. Why is 2 the only even prime? 2

13. Find $(x + 3)(x + 5)$. 2

14. Find $(x-4)(x+4)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain the test for divisibility by 6 using 2 and 3. Test 4728. 3

16. Expand and simplify $(x+3)(x+4)$. Verify for $x=1$. 3

17. Apply Kaprekar's routine to 3524 until it reaches 6174. 3

OR

Find the missing digit: $3\square52$ is divisible by 11. 3

18. Expand and simplify $(x+5)(x+7)$. Verify for $x=1$. 3

OR

Expand and simplify $(x+2)(x+9)$. Verify for $x=1$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Prove that the product of three consecutive integers is divisible by 6. 5
(ii) Verify for 7×8×9.

OR

(i) A number has digits 4, 7 and 1 in some order. Form the largest and smallest numbers and subtract. 5
(ii) Repeat the process and reach 495.

20. (i) Show that $(a+b)^2=(a-b)^2+4ab$. 5
(ii) Use it to find $(15+9)^2-(15-9)^2$.

OR

(i) State and illustrate the commutative, associative and distributive properties for multiplication. 5
(ii) Which property is used in $18\times25\times4$?

END OF QUESTION PAPER — Total: 40 Marks

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